An Analysis of Circuit Fidelity and Error Accumulation in IBM Quantum Computers

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***Abstract*—Quantum computers in the Noisy Intermediate-Scale Quantum (NISQ) era are heavily constrained by environmental noise and decoherence. While daily automated calibrations track static error baselines, they fail to track the constant intra-day fluctuations that decrease measurement accuracy. In this study, we conducted a 24-hour diagnostic benchmark on a contiguous three-qubit sub-graph of the 156-qubit `ibm_marrakesh` heavy-hex processor from IBM. Every four hours, we executed a suite of 28 unique circuits across six experimental categories at 8,192 shots per circuit. Single-qubit operations showed high stability, maintaining measurement fidelities above 99.6% across depths up to 1,024 Pauli-X gates. Conversely, multi-qubit entanglement and excited-state measurements showed much weaker stability, noticeably during a localized thermal event that caused 3-qubit GHZ state fidelity to drop from 97.3% to 93.6% along with routing errors. Our measured error exceeded the theoretical error up to 11.8 times greater, demonstrating that these fluctuations come from physical calibration drift rather than statistical artifacts. Aside from measuring the error loss contributors, these findings show that static, daily calibrations are insufficient for characterizing NISQ hardware stability.**

I. Background

Unlike classical computers that process information in static binary states of 0 or 1, quantum computers use principles of quantum mechanics to process information using quantum bits, or qubits. Qubits utilize the property of superposition, where the qubit can exist in both the $|0\rangle$ and $|1\rangle$ state simultaneously [1]. Quantum logic is executed by coupling these qubits together to create entanglement, where the states of multiple qubits are linked across space [2].

However, current hardware operates within the Noisy Intermediate-Scale Quantum (NISQ) era, meaning that qubits are highly susceptible to environmental interference, thermal fluctuations, and temporal drift [3]. These disruptions, characterized by thermal relaxation (T_1) and dephasing (T_2), cause quantum states to decay prematurely [4]. Thermal relaxation occurs when an excited qubit loses energy to its environment, collapsing from the $|1\rangle$ state back to the $|0\rangle$ ground state, a problem we can easily see during state measurement. Dephasing happens when environmental fluctuations, such as stray magnetic fields, scramble the phase relationship of a superposition state without causing energy loss. Beyond environmental decoherence, operations can gather noise due to control errors and spatial cross-talk, where driving a gate on one qubit unintentionally disturbs its physical neighbors on the heavy-hex lattice.

This noise is a major problem in quantum computing, because it quickly destroys delicate quantum information, limiting how complex programs can be and whether the results can be trusted. Periodic calibrations attempt to provide information to assist with modeling the noise. However, this noise fluctuates continuously, limiting the efficacy of these calibrations. In order to be able to design effective error-correction software or identify the weakest parts of the quantum system, we need a clear map of the processor's instability.

In this study, we measured the temporal stability of a real quantum processor over a full 24-hour cycle by executing a suite of 28 diagnostic circuits on a three-qubit sub-graph within the IBM processor.

II. Testbed: IBM Quantum Computer

The computer we used for this study was an IBM Quantum heavy-hex processor [5], specifically, the backend named `ibm_marrakesh` with 332 physical qubits of which 156 are programmable. The backend uses superconducting transmon technology. Their artificial atoms are constructed using Josephson junctions and operate at temperatures near absolute zero within dilution refrigerators. This extremely cold environment is necessary to maintain the microwave energy states required for quantum computation and to protect the rest of the system from unwanted thermal interference [6], [7].

The physical qubits on `ibm_marrakesh` are arranged in a heavy-hex lattice topology (see Figure 1). This hexagonal architectural design is intentionally utilized by IBM to mitigate spatial cross-talk and reduce frequency collisions between neighboring qubits [5]. However, this design strictly limits physical connections; each qubit is only directly coupled to two or three adjacent qubits, and as a result, executing logic between non-adjacent qubits requires routing entanglement across nodes which exposes the algorithm to the specific edge noise and path distances that we measured.

The entire experimental suite was programmed and deployed using Qiskit, an open-source Python package for quantum computing [8]. Qiskit served as the primary interface for building the circuits, managing the 8192-shot measurement parameters, and executing operations on specific contiguous qubits (0, 1, and 2). Qiskit Runtime has a built-in error-mitigation system. Our software environment bypassed this system by setting the Qiskit resilience and optimization levels to zero. The execution jobs were then passed one after the

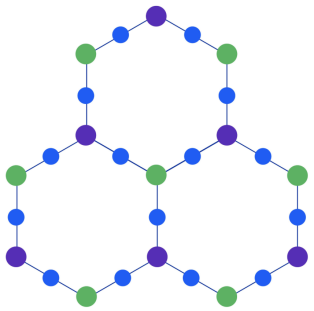


Fig. 1. IBM heavy-hex design [5].

other to the physical process via the IBM Quantum Cloud API.

III. Methodology

Our goal was to try to explore and analyze different sources of quantum noise by running six unique experiments. To capture temporal drift, we ran a scheduler consisting of these six experiments, running them periodically every four hours across a complete 24-hour cycle. Each circuit was measured using 8,192 shots, ensuring that any observed errors represented genuine physical hardware drift rather than statistical artifacts. Table I provides the details of the experiments that we ran.

IV. Limitations

IBM Quantum provides a maximum of 10 minutes of free runtime per month, and these experiments utilized nearly all of it. Running more in-depth experiments would require exceeding this limit and switching to a pay-as-you-go plan at a rate of $1.60 per second of usage. For our original experimental scope, this would have cost over six thousand dollars, significantly exceeding our budget. As a result, we refined and reduced the experiment to reach the final iteration, trading off cost for a smaller but still statistically significant data footprint.

Restricting the benchmark to a 3-qubit adjacent chain was necessary for isolating specific readouts, but it evaluated less than 2% of the total 156-qubit processor. Additionally, hardware performance and overall fidelity and error rates may vary across different physical regions of the heavy-hex lattice. Therefore, the behavior observed on qubits 0, 1, and 2 cannot be generalized to the entire chip.

Sampling every 4 hours provided a clear view of 24-hour drift, but it missed high-frequency deviations such as minute-by-minute phase instabilities or rapid microwave power shifts. The 24-hour window is also insufficient for observing broader trends over multiple days or weeks, such as permanent calibration shifts or cumulative hardware degradation over time.

Finally, the precise 4-hour intervals targeted by our automated script were subject to variance due to IBM Quantum's shared job queue. Minor timing offsets were introduced depending on queue congestion at the time of each execution, slightly reducing the exact temporal consistency between runs.

V. Results

A. Depth Scaling

This experiment evaluated single-qubit gate error accumulation by executing repeated, sequential chains of 2, 128, 512, and 1024 Pauli-X gates on each Qubit 0, 1, and 2. Figure 2 shows the reference circuit with depth of two Pauli-X gates. The full test scaled up to 1024 sequential X gates on Qubits 0, 1, and 2. Applying an even number of bit-flip operations theoretically should return the qubit to its initial $|0\rangle$ ground state, allowing the test to measure how deep microwave pulse sequences accumulate errors over time without thermal relaxation completely messing up the state by seeing how many $|1\rangle$ states show up after the operations. Figure 3 shows the results.

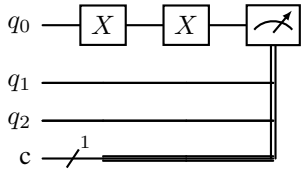


Fig. 2. Depth scaling experiment circuit on Qubit 0 executing two sequential Pauli-X gates. The full experiment extended the X gates up to depth 1024.

All three qubits retained near-perfect baselines across the 24-hour cycle. Qubit 0 was the most stable, averaging a fidelity of 99.92% at 1024 gates with a maximum variance of only 0.15%. Qubit 2 was similarly dominant, averaging 99.86% fidelity at 1024 gates. Qubit 1 was the weakest qubit for deep circuits, but still averaged a strong 99.61% fidelity at maximum depth. There seems to be no apparent relationship between depth and fidelity.

B. Topology Superposition

Here we are applying a single Hadamard gate to each physical qubit, ideally projecting the qubit into an equal superposition of ground and excited states:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Figure 4 shows one of the circuits with a Hadamard gate on Qubit 0. The full test also tested the Hadamard gate on Qubits 1 and 2. By calculating the absolute deviation of the measured

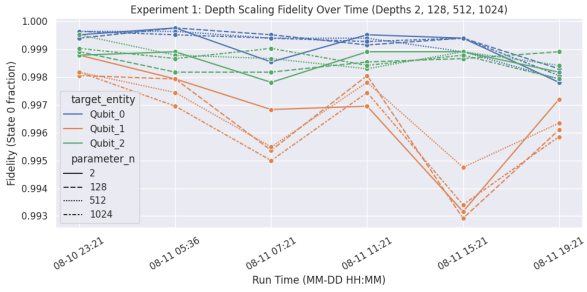


Fig. 3. Depth Scaling Fidelity Over Time (Depths 2, 128, 512, 1024)

TABLE I

Experiment	Target Focus	Circuits	Description
Depth Scaling	Error Accumulation	12	Executing continuous sequences of up to 1,024 Pauli-X gates.
Topology Superposition	Single-Qubit Coherence	3	Testing single-qubit coherence by applying a Hadamard gate to measure the hardware's deviation from an ideal 50/50 quantum superposition.
Temporal Bell Pairs	2-Qubit Entanglement	2	Generates Bell states across adjacent physical connections (Q_0-Q_1 and Q_1-Q_2) to measure baseline entanglement fidelity.
GHZ Scaling	Multi-Qubit Entanglement	3	Testing multi-qubit entanglement by scaling Greenberger-Horne-Zeilinger (GHZ) states across 1, 2, and 3 adjacent qubits.
SPAM Readout	Asymmetric Readout Error	6	Isolates state preparation and measurement (SPAM) asymmetries, specifically tracking thermal relaxation when reading the excited $\|1\rangle$ state.
Path Distance	Routing Fidelity	2	Testing how effectively entanglement is maintained when forced across longer physical distances on the chip.
Total	**Temporal Drift**	**28**	**Captures complete snapshot of hardware health every 4 hours.**

output counts from an ideal 50/50 split across 8,192 shots, we can see subtle phase drift across the processor. Figure 5 shows the results.

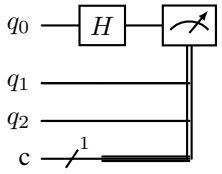


Fig. 4. Topology superposition experiment circuit applying a single Hadamard gate to Qubit 0.

A perfect result here is 0.0. The graph displays the absolute deviation from this ideal state. We can see that Qubit 2 showed the closest adherence to ideal superposition, averaging a minor deviation of just 0.005. Qubit 0 and Qubit 1 experienced much larger thermal fluctuations during the mid-day runs (7:21-11:21 UTC), with Qubit 0's deviation spiking to 0.043.

C. Temporal Bell Pairs (Entanglement)

This experiment measured two-qubit entanglement fidelity across physically adjacent nodes on the heavy-hex lattice by generating maximally entangled Bell states on the adjacent qubit pairs (Q_0-Q_1 and Q_1-Q_2). This was done using a single Hadamard gate followed by a $CNOT$ gate, then allowing us to isolate native two-qubit gate error rates across the specific routing edges. Figure 6 shows the reference circuit with Qubits 0 and 1 entangled. The full test tested entanglement between Qubits 1 and 2 as well. Figure 7 shows the results.

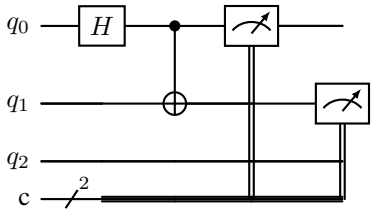


Fig. 6. Temporal Bell Pairs experiment circuit generating entanglement between Qubit 0 and Qubit 1.

The Q_1-Q_2 connection is clearly the preferred routing edge on the heavy-hex lattice, averaging 98.48% fidelity and remaining above 97.5% for the entire 24 hours, while the Q_0-Q_1 connection is significantly noisier, averaging just 97.15%

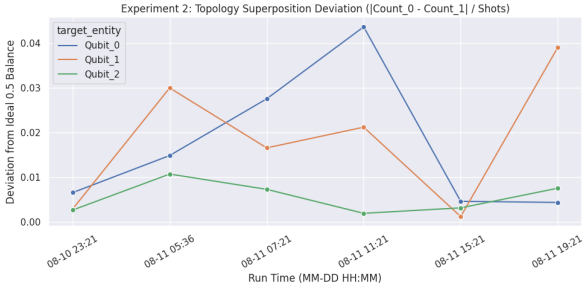


Fig. 5. Topology Superposition Deviation ($\frac{|Count_0 - Count_1|}{Shots}$)

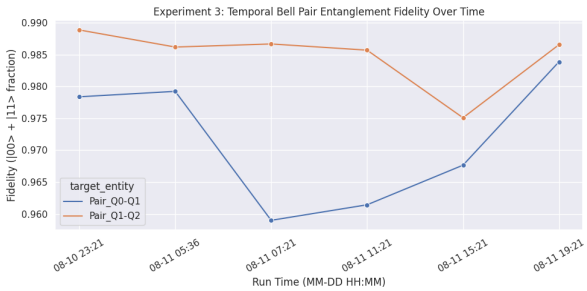


Fig. 7. Temporal Bell Pair Entanglement Fidelity Over Time

and even dipping to a low of 95.89

D. GHZ Scaling (Multi-Qubit Engtanglement)

Here we similarly measured how entanglement fidelity decays but with expanding the register size by constructing multi-qubit Greenberger-Horne-Zeilinger (GHZ) states across 1, 2, and 3 contiguous physical qubits. Specifically, we implemented cascading single-qubit entanglements using successive $CNOT$ gates. Figure 8 shows the 3-qubit circuit measuring the GHZ state. Figure 9 shows the results of the experiment.

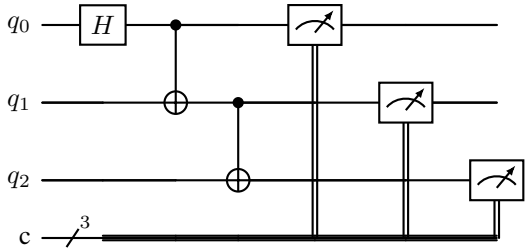


Fig. 8. GHZ Scaling experiment circuit generating a 3-qubit entangled state.

The 1-qubit baseline held at a flat 1.00. Once a second qubit was introduced, the mean fidelity immediately dropped to 97.31%. Adding the third qubit dragged the average down further to 96.3%. More notably, the 3-qubit state was highly vulnerable to a mid-day thermal event at 07:21 UTC, crashing to an absolute minimum of 93.66%. We can clearly see the inverse relationship between fidelity and entangled qubit count.

E. SPAM Readout (Measurement Asymmetry)

Here we isolated State Preparation and Measurement (SPAM) errors by preparing each qubit in both the ground state ($|0\rangle$) and the excited state ($|1\rangle$) immediately before measurement. Figure 10 shows the circuits we used for preparing Qubit 0 in ground and excited state, respectively. The full test included this sequence on Qubits 1 and 2 as well. By measuring the fidelity of prepared ground and prepared excited states, we are able to see state-dependent measurement asymmetries, specifically seeing the impact of thermal relaxation (T_1 decay) occurring during the readout pulse. Figure 11 shows the results.

This data shows that measurement errors on IBM's hardware are highly asymmetric. All three qubits measure the ground $|0\rangle$

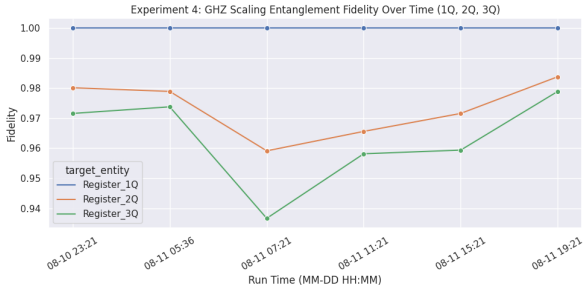


Fig. 9. GHZ Scaling Entanglement Fidelity Over Time

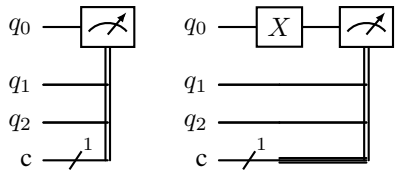


Fig. 10. State Preparation and Measurement (SPAM) circuits for Qubit 0. The left circuit measures the default $|0\rangle$ ground state, while the right circuit applies a Pauli-X gate to prepare and measure the $|1\rangle$ excited state.

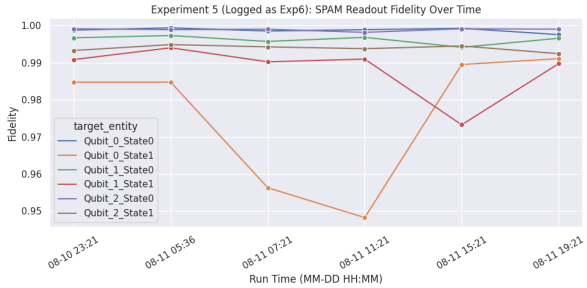


Fig. 11. SPAM Readout Fidelity Over Time

state almost perfectly, with averages around 99.6% to 99.8% across all runs. The excited $|1\rangle$ state was far more unstable due to thermal relaxation during the readouts. Qubit 0 is extremely effective in this context, with it being an outlier and having the lowest fidelity readout by more than 2% greater than the next closest readout.

F. Path Distance Routing

This experiment evaluated topology-dependent errors by comparing two-qubit entanglement fidelity generated across a direct 1-hop physical link (Q_0 to Q_1) against an extended 2-hop path (Q_0 to Q_2 routed through Q_1), which requires an extra $CNOT$ gate. This can be visualized using the heavy-hex lattice design of the IBM quantum chip in Figure 12 and through the circuits shown on Figure 13. Here, we see how the Q_0 to Q_2 path is forced to cascade through the intermediate Q_1 node because there is no direct diagonal physical connection between Q_0 and Q_2. Figure 14 shows the results.

The direct path averaged 97.15% fidelity while the longer path averaged 97.43% fidelity. Mathematically, the longer path should be noisier, but it occasionally outperformed the shorter path, suggesting that the noise on the Q_0-Q_1 edge is severe and much more significant than the noise on the second half of the 2-hop path.

G. Overall 24-hour Drift

Over the complete 24-hour observation window, the experiments showed a stark difference between single-qubit gate executions and more complex quantum operations. Single-qubit operations showed exceptional baseline fidelity, as evident by the Depth Scaling experiment, applying up to 1,024 sequential Pauli-X gates yielded stable fidelities of over 99.6% across all timeframes. However, operations based on multi-qubit spatial entanglement and excited-state holds showed significant

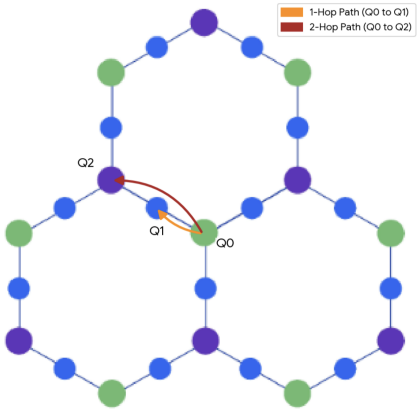


Fig. 12. Routing on Heavy-Hex Lattice (1-hop vs. 2-hop)

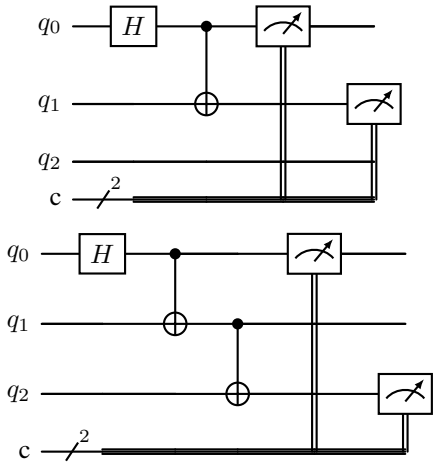


Fig. 13. Path Distance routing fidelity circuits. The top circuit evaluates a standard 1-hop entanglement path directly between adjacent qubits (Q_0 and Q_1). The bottom circuit evaluates a 2-hop path by extending the entanglement to Q_2 before measuring the terminal endpoints (Q_0 and Q_2).

instability over time. The hardware exhibited distinct, fleeting phases of instability. The initial nighttime runs showed a high-fidelity baseline, but soon suffered a mid-day "thermal event," subsequently recovering to its initial baseline by the end of the 24-hour cycle. Importantly, the backend's reported T_1 and T_2 times remained completely flat and failed to register any of these shifts.

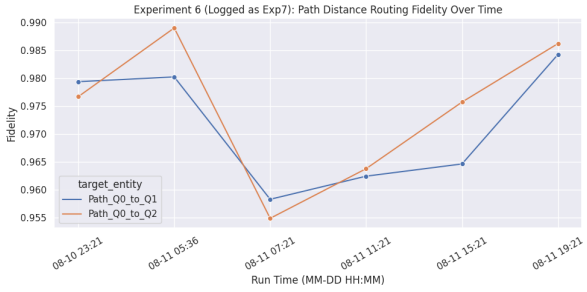


Fig. 14. Path Distance Routing Fidelity Over Time

The "thermal event" was the most significant moment of deviance of our experiment occurring, during the 07:21 UTC run, where we can see a pronounced, localized hardware anomaly. This event disturbed almost every experiment we ran, causing fidelity to dip greatly. At 07:21, the hardware temporarily lost its ability to reliably maintain spatial correlations. The 3-qubit GHZ state went from a previous baseline fidelity of $\sim$97.3% suffered a sudden drop down to 93.6%. Concurrently, the 2-qubit Bell pair fidelity on the Q_0-Q_1 edge dropped from 97.9% to 95.8%. In the depth scaling test (Exp 1), Qubit 1 experienced a noticeable dip as well, which combined with the collapsing entanglement suggests that the hardware disruption at 07:21 was physically localized near Qubit 1 or the microwave lines feeding it. While Qubit 0 and Qubit 2 maintained exceptional single-qubit depth fidelities (>99.8%) during this window, the disruption specifically decreased Qubit 1's individual performance and any entanglement logic ($CNOT$s) that relied on it.

TABLE II
EXPECTED STANDARD DEVIATION VS. MEASURED TEMPORAL DRIFT.

Experiment	Expected Error (Shot Noise)	Actual Error (Measured Drift)	Difference (Actual vs. Expected)
Depth Scaling	$\pm$ 0.048%	$\pm$ 0.156%	3.2× larger
Topology Superposition	$\pm$ 0.552%	$\pm$ 1.332%	2.4× larger
Temporal Bell Pairs	$\pm$ 0.161%	$\pm$ 1.037%	6.4× larger
GHZ Scaling	$\pm$ 0.159%	$\pm$ 1.878%	11.8× larger
SPAM Readout	$\pm$ 0.099%	$\pm$ 1.117%	11.3× larger
Path Distance	$\pm$ 0.179%	$\pm$ 1.163%	6.5× larger

Table II shows the expected standard deviation of each experiment based on sample size (8192 shots per circuit; circuit count varies per experiment) contrasted against the actual standard deviation of the hardware across the 24 hours. We used the formula for standard error (p is averaged observed fidelity and $N = 8192$) to get the error rates. Because the measured drift was consistently between 2.4 and 11.8 times larger than the expected statistical error, the data proved that the dips in fidelity we observed represent genuine changes in the quantum processor's calibration state rather than random sampling artifacts. Notably, the GHZ Scaling and SPAM experiments exhibited the highest actual-to-expected-error ratios, highlighting them as the operations most vulnerable to intra-day physical hardware degradation.

VI. CONCLUSION

In this paper, we evaluated how physical hardware noise on the IBM `ibm_marrakesh` heavy-hex quantum processor fluctuates across a complete 24-hour cycle. We found a stark contrast between the results of basic operations and of complex quantum states. The single-qubit operations had exceptional baseline stability, while multi-qubit entanglement and excited-state readouts were much less stable and susceptible to temporal physical drift and thermal relaxation. Most notably, we isolated a localized "thermal event" at 07:21 UTC. This

anomaly temporarily hindered the processor's ability to maintain certain spatial correlations, causing the 3-qubit GHZ state fidelity to drop down to 93.6% and heavily degrading the Q_0-Q_1 routing edge. Importantly, the backend's reported baseline T_1 and T_2 measurements stayed flat and completely failed to register this shift. By contrasting our measured temporal standard deviation against the theoretical statistical shot noise floor (which reached up to 11.8 times larger than the expected statistical variance), we mathematically confirmed that the "event" presented genuine hardware degradation rather than random sampling artifacts. It is highly unlikely that we got "lucky" in detecting the thermal event. Rather, our detection of it suggests that these such events happen regularly.

This suggests that IBM's once-a-day calibrations are insufficient for reliably characterizing NISQ hardware. Because thermal events like this happen dynamically throughout the day and are highly localized to specific regions and routing edges, researchers cannot rely on past calibrations for current algorithms. Instead, they must implement continuous benchmarking to increase the accuracy of the calibrations in real-time.

Interesting future areas of study in this topic include investigating if thermal events ripple across larger regions of the 156-qubit lattice, not just on three qubits. Additionally, investigating methods for continuous benchmarking that allow for efficient circuit execution and error correction.

Code and data of this study is online at https://github.com/mlonneamir/qnoise.

VII. AI Usage Statement

In this paper, we used Google Gemini to assist with creating the Qiskit simlulator code. We also used AI assistance to generate the LaTeX code for the circuit diagrams and the graphs.

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